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A Green, Reagent-Saving Analytical Strategy for Equivalence Point Determination in Potentiometric Titrations via Gran and Schwartz Linearization†

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Potentiometric titrations remain a cornerstone of analytical chemistry however require titrant (reagent) volumes in excess due to the need for full-curve exploration to locate the equivalence point (EQP), contributing to unnecessary waste and contradicting green chemistry principles. Here, we introduce a sustainable two-stage analytical strategy for accurate and reagent-efficient EQP determination. In the method development stage, an automated titration with variable titrant increment size identifies the pre-equivalence linear region via Gran and Schwartz linearization methods. An optimal earliest acceptable point volume is then automatically selected according to user-defined criteria, i.e., coefficient of determination R^2, volume standard error, and agreement with the full-dataset EQP. In the method validation and method application stages, titrations are performed only up to this optimized point, enabling precise EQP extrapolation from significantly reduced data. Applied to the aqueous titration of tris(hydroxymethyl)aminomethane (TRIS) with HCl and potassium hydrogen phthalate (KHP) with NaOH (n=6 replicates), this approach yields EQP values with excellent repeatability while reducing titrant consumption by 10–20% per analysis (depending to the applied threshold criteria). Implemented in the open-source Python tool GranTED (available at `https://github.com/sgiani95/GranTED`), this protocol offers a practical, low-waste solution for routine laboratory and educational potentiometric analyses, directly supporting Principles 1 and 5 of Green Chemistry.

Introduction

Titration is a quantitative chemical analysis method for determining the concentration of a substance (the analyte) in solution. A reagent of known concentration (the titrant) is gradually added to the analyte until the reaction is complete. The endpoint is detected using a color indication or instrumental measurements (i.e. potential E or pH). Traditional methods rely on identifying the inflection point in E or pH versus volume (V) curves, but these often suffer from low accuracy in systems with small potential breaks, such as weak electrolytes or asymmetric curves. To address this, Gran introduced differential plotting of $\Delta V/\Delta E$ or $\Delta V/\Delta \mathrm{pH}$ versus V in 1950,[1] yielding near-linear segments intersecting at the equivalence point (EQP). In 1952, Gran refined this with antilogarithmic transformations (e.g., $(V_0+V)\times 10^{-\mathrm{pH}}$ versus V),[2] producing straight lines for easier extrapolation. Schwartz further advanced Gran methodology in 1987 by incorporating corrections for significant $[H^+]$ or $[OH-]$ perturbing effects, enabling analysis of challenging cases.[3] A comprehensive study on these evaluation techniques by means of simulations and experimental tests has been presented 2018 by Martin et al.[4]

Despite these methodological improvements, potentiometric titrations remain resource-intensive, requiring additional curve acquisitions after the EQP that consume excess titrants, generating unnecessary waste contrary to green chemistry principles (waste prevention and safer solvents).[5] In this work, we present a sustainable two-stage analytical strategy that minimises solvent use while leveraging Gran and Schwartz linearizations for accurate and precise EQP determination.

The protocol involves (i) an initial automated titration with fixed and variable titrant increments until post-EQP, to identify the pre-EQP linear region for estimation. An optimal earliest acceptable point volume is then automatically determined according up to three user-tunable criteria (coefficient of determination R^2, volume uncertainty and agreement with the full-dataset EQP); and (ii) method validation and method application stages, in which subsequent titrations use calculation and titrant dispensing only up to this optimized point, enabling precise EQP extrapolation from significantly reduced data. Applied to two case systems, tris(hydroxymethyl)aminomethane (TRIS) –HCl and potassium hydrogen phthalate (KHP) with NaOH titrations (n=6 repli-

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cates), the approach achieves excellent repeatability while reducing titrant consumption. This strategy, implemented in the open-source Python tool GranTED, directly supports green chemistry Principles 1, and 5 and offers a practical, tunable solution for routine laboratory and educational potentiometric analyses.

Open-source implementation in Python (GranTED, available at `https://github.com/sgiani95/GranTED`) facilitates its adoption.

Experimental Section

Materials and Instrumentation

Tris(hydroxymethyl)aminomethane (TRIS, 99.97%, Merck Sigma-Aldrich) and potassium hydrogen phthalate (KHP, >99.5%, Merck Sigma-Aldrich) were titrated with hydrochloric acid (HCl) and sodium hydroxyde (NaOH) respectively (both 0.1 M, Titripac Merck Millipore). Aqueous solutions were prepared in deionized water (18.2 M$M\Omega \cdot cm$). Potentiometric measurements have been performed by means of a Mettler-Toledo T5 automated titrator equipped with a combined pH glass electrode (DGi111-SC). All experiments were conducted at room temperature.

Protocols

The analytical protocol comprises two distinct stages.

Method development stage: 5 mL of 0.1 M TRIS (or 0.1 M KHP) was diluted with 60 mL deionized water and titrated with 0.1 M HCl (or 0.1 M NaOH) using fixed 0.05 mL increments until past the equivalence point (post-EQP). The collected data were processed using both the conventional inflection-point method and the Schwartz-modified Gran plots where $(V+k)\times 10^{-\mathrm{pH}}$ is plotted against V (V = the dispensed titrant volume, k = optimization parameter). LThe pre-equivalence linear region was identified, and the *optimal earliest acceptable point volume* was automatically identified according to three user-tunable criteria: $R^2 \geq 0.9995$, standard error of the extrapolated EQP ≤ 0.01 mL, and agreement with the full-dataset EQP within ± 0.01 mL.

Method application stage: Six replicate titrations were performed until past the equivalence point using the same 0.05 mL increment protocol. EQP were determined both by conventional inflection-point method and by linear extrapolation from the Gran–Schwartz plot for direct comparison and method validation. Routine titrations of unknown samples were subsequently carried out by dispensing titrant only up to the previously determined optimal earliest acceptable point volume using the same fixed-increment protocol.

Data Analysis

All titration data were analysed using the open-source Python package GranTED (`https://github.com/sgiani95/GranTED`). Results are expressed as mean ± standard deviaiton (n = 6). Solvent savings were calculated as the differences dispensed titrant volume between the full conventional titration (continued past the equivalence point) and the titrant at the optimal earliest acceptable point volume (Gran–Schwartz EQP). Detailed equations, raw data, and algorithm pseudocode are available in the Electronic Supporting Information (ESI).

Results and Discussion

Method Development: Optimization of the Earliest Acceptable Point Volume and Equivalence Point Determination by Gran–Schwartz Linearization

The sustainable analytical strategy begins with a single conventional titration performed to establish the baseline pH–V curve. For the titration of 5 mL of 0.1 M TRIS (diluted in 60 mL deionized water) with 0.100 M HCl, the resulting sigmoidal curve (Figure 1a) showed the expected behaviour of a weak base–strong acid system ($pK_a(\mathrm{TRIS}) \approx 8.1$). Application of the Schwartz-modified Gran transformation, $(V+k)\times 10^{-\mathrm{pH}}$ versus titrant volume (Figure 1b), successfully linearized the pre-equivalence region. The linear segment (shaded) spans approximately 2.5–5.0 mL.

Determination from the pre-EQP linear fit provides the equivalence point without requiring full post-equivalence data. In the initial titration the Schwartz plot (Figure 1b) intersects the V-axis at 5.002 mL (theoretical EQP), with a standard error of ± 0.005 mL propagated from linear regression. This aligns with Gran's theory for transforming curved branches into straight lines, enhanced by Schwartz's electroneutrality corrections to account for $[H^+]$ dominance in the buffer region. The optimal earliest acceptable point volume ensures that only the most reliable pre-EQP data are used for extrapolation.

Automated linear-region identification combined with backward data trimming revealed an *optimal earliest acceptable point volume* of 4.40 mL (Table 1). This value satisfied all three user-tunable criteria ($R^2 \geq 0.9995$, standard error of the extrapolated EQP ≤ 0.01 mL, and agreement with the full-dataset EQP within ± 0.01 mL) and was adopted as the stopping criterion for all subsequent titrations. The chosen point lies safely before the equivalence point, enabling reliable extrapolation while substantially reducing titrant consumption, enabling reliable extrapolation akin to Gran's functions intersections. This approach substantially reduces titrant consumption compared to conventional full titrations while maintaining the ability to determine the equivalence point through Gran–Schwartz linear extrapolation.

Table 1 Gran–Schwartz results for the TRIS/HCl system in the vicinity of the optimal earliest acceptable point volume (4.40 mL, in bold). Full EQP corresponds to 5.007 mL. The selected volume meets all acceptance criteria (e.g. deviation from full EQP smaller than 0.01 mL) with excellent linearity and low uncertainty. Except for R^2, all values in mL

Dispensed volume	EQP	Standard Error	R^2	Deviation from full EQP
4.25	4.980	0.010	0.9987	-0.027
4.30	4.987	0.010	0.9988	-0.020
4.35	4.989	0.009	0.9989	-0.018
4.40	**5.002**	**0.005**	**0.9997**	**-0.005**
4.45	5.002	0.005	0.9997	-0.005
4.50	5.007	0.004	0.9998	0.000
4.55	5.006	0.003	0.9998	-0.001

Subsequent titrations performed up to the optimal earliest acceptable point volume (Figure 1c) confirmed the strategy's accuracy. The corresponding Schwartz plot (Figure 1d) mir-

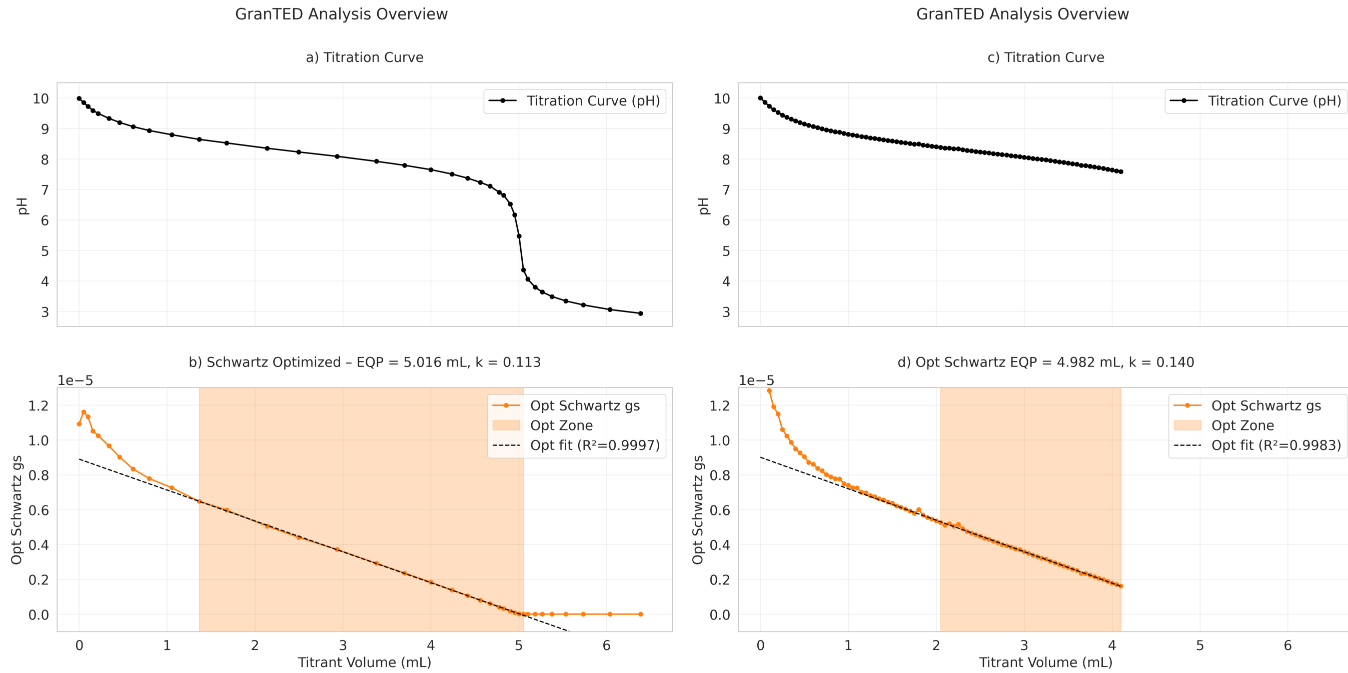

Fig. 1 Overview of the green analytical strategy for equivalence point (EQP) determination in potentiometric titrations, demonstrated for TRIS (0.1 M) titrated with HCl (0.1 M). Left: a) Raw pH vs. titrant volume (V) curve from the initial titration (full run to post-EQP). b) Schwartz–Gran modified plot ($V \times 10^{-\mathrm{pH}}$ vs. V), showing the straight pre-EQP line extrapolated to the EQP (dashed line; intercept at 5.020 mL). Right: c) Titration performed up to the optimal earliest acceptable point volume. d) Corresponding Schwartz–Green modified plot confirming EQP extrapolation (intercept at 5.002 mL) with reduced dispensed volume.

rors the linearisation and extrapolates to an EQP of 5.002 mL (±0.005 mL), matching the conventional inflection-point value within experimental error.

Method Validation and Performance: Comparison with Conventional Inflection Point Method

Subsequent titrations performed up to the optimal earliest acceptable point volume confirmed the strategy's accuracy for both systems studied. For the TRIS/HCl (aq) and KHP/NaOH (aq) titrations, the corresponding Schwartz-modified Gran plots gives determined EQP values that matched the conventional inflection-point results within threshold for the stopping criterion (Table 2).

Across $n = 6$ replicates per method and per system, repeatability was excellent. The standard deviation obtained with the optimised early-stop protocol was consistent with the chosen threshold for stopping, demonstrating robustness against volume variability and validating the practical utility of the tunable early-stop criterion.

Table 2 Comparison of equivalence point (EQP) determination by conventional inflection-point method (post-EQP) and Gran–Schwartz linear extrapolation stopped at the optimal earliest acceptable point volume corresponding to the indicated deviation from full EQP. Results are mean ± SD ($n = 6$ replicates). All values are in mL

		Gran–Schwartz		
	post-EQP	0.01	0.02	0.05
TRIS/HCl	5.015 ± 0.001	4.990 ± 0.010	5.005 ± 0.015	4.992 ± 0.064
KHP/NaOH	5.032 ± 0.001	5.045 ± 0.003	5.051 ± 0.015	5.066 ± 0.041

Green Chemistry Metrics and Sustainability Assessment

Early termination at the optimal earliest acceptable point volume yields systematic titrant savings. As shown in Figure 2, stopping titrations according to the different acceptance criteria (0.01, 0.02 and 0.05 mL difference from the full-dataset EQP) results in savings of approximately 10% up to 30% versus the EQP and approximately 5% more per case considering the full endpoint of the titration (0.5 mL post-EQP) per analysis for both the TRIS/HCl and KHP/NaOH systems. This corresponds to corresponding to a bare minimum of 12.07% and 10.80% reductions respectively. For routine laboratories performing multiple replicates, this translates into substantial waste minimisation.

The approach directly embodies Green Chemistry Principle 1 (prevention) by avoiding unnecessary full runs and Principle 5 (safer/less solvents) through reduced HCl or NaOH exposure by lowering waste production.[5] Compared with traditional methods that require titrant in excess, the protocol decreases resource consumption without compromising accuracy or precision. Implemented in the open-source Python tool GranTED, it automates method development, validation, and routine application, promoting the adoption of more sustainable practices in educational, analytical laboratory, and industrial contexts.

Conclusions

In conclusion, we have developed a green analytical strategy for potentiometric equivalence point (EQP) determination that leverages Gran–Schwartz linearizations to enable titration stopping at the *optimal earliest acceptable point volume*, automatically se-

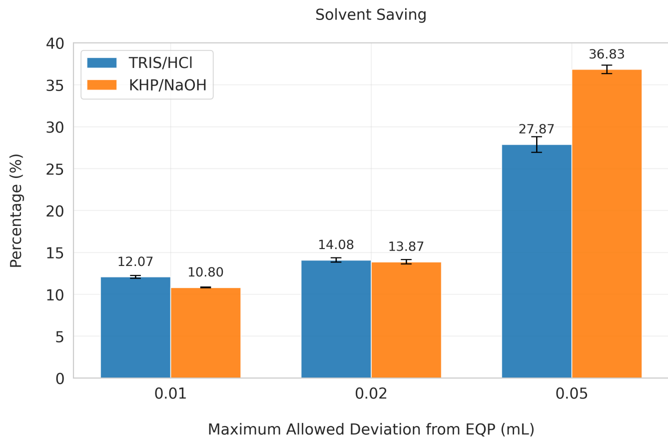

Fig. 2 Solvent savings achieved by stopping at the optimal earliest acceptable point volume using different acceptance criteria (0.01, 0.02 and 0.05 mL difference from the full-dataset EQP). Results are shown for both TRIS/HCl and KHP/NaOH systems ($n = 6$ replicates each) and compared against the equivalence point and full titration endpoint

lected according to user-defined criteria (most critically, deviation from the full-dataset EQP smaller than a chosen threshold).

Applied to both TRIS/HCl and KHP/NaOH aqueous titrations, the protocol delivers highly repeatable EQPs (SD ranging from ± 0.001 mL to circa ± 0.050 mL, $n = 6$, depending on the applied threshold) while systematically reducing solvent and titrant consumption by approximately 10–30 %. This approach aligns with Green Chemistry Principles 1 (prevention) and 5 (safer solvents) through substantial waste minimisation without compromising accuracy or precision.

Furthermore, the same tunable criteria open the possibility for an *online, direct forward approach* in which the titration is automatically terminated on-the-fly, during the titration itself, in a sample-driven manner, as soon as the Gran–Schwartz linear fit results satisfy the user-defined thresholds. This real-time adaptive termination would lead to the same solvent-saving benefits while enabling fully dynamic, adaptive titrations.

Implemented in the open-source Python tool GranTED, the analytical strategy offers a practical and sustainable alternative for routine laboratory and educational applications, with straightforward extension to complex mixtures, environmental samples, and automated high-throughput systems. These results demonstrate that reliable Gran–Schwartz extrapolation can be achieved using only a fraction of the titrant typically consumed in conventional potentiometric titrations, without compromising analytical accuracy or precision.

Author Contributions

S.G. and C.A.D.C. conceptualized the study, developed the methodology, performed the investigation and formal analysis, curated the data, visualized the results, and wrote the original draft. S.G. developed the software. C.A.D.C. carried out the experiments.

Conflicts of interest

S.G. declares no conflicts of interest. C.A.D.C. is currently working for a manufacturer of laboratory analytical instrumentation (Mettler-Toledo Technologies GmbH).

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Data availability

The datasets supporting this article are available in the Electronic Supporting Information (ESI) or from the corresponding authors upon reasonable request.

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